34th Annual INCOSE international symposium
hybrid event
Dublin, Ireland
July 2 - 6, 2024

AI Systems Modeling Enhancer (AI-SME): Initial Investigations into a ChatGPT-enabled MBSE Modeling Assistant

Brian Johns
United States Air force Academy.
Colorado Springs, CO USA
brian.johns@afacademy.af.edu

Kristina Carroll
Studio SE, Ltd
Larkspur, CO USA
kristina.carroll@studiose.design

Casey Medina
Studio SE, Ltd.
Larkspur, CO USA
casey.medina @studiose.design

Rae Lewark
Studio SE, Ltd.
Larkspur, CO USA
rae.lewark@studiose.design

James Walliser
United States Air Force Academy
Colorado Springs, CO USA
casey.medina@studiose.design

Abstract. As artificial intelligence (AI) becomes integral across industries, there is a growing opportunity to transform the generation of models for systems engineering. This research investigates the integration of OpenAI's GPT-4 Turbo into CATIA Magic for Model-Based Systems Engineering (MBSE), resulting in the creation of AI Systems Modeling Enhancer (AI-SME). This study explores the comparison between models generated by AI, specifically OpenAI's GPT, and those crafted by human systems engineers. While recognizing challenges in AI-generated models, this research underscores the potential of AI assistants to enhance the speed and accuracy of SysML model creation. Results demonstrate AI-SME's successes in generating requirements, block definition diagrams, and internal block diagrams. Despite identified limitations such as redundancy and lack of cohesiveness in AI-generated models, the study concludes that AI-SME represents a notable advancement in AI-assisted systems engineering.

Keywords. AI, AI-integration, ChatGPT, GPT-4, MBSE, SysML, Systems Modeling, Modeling Assistant.

Introduction

The advent of large language models (LLMs) such as OpenAI's family of GPT models has created the opportunity for significant disruption across multiple domains driving industry professionals to investigate potential use cases. Given ChatGPT's relatively brief existence (13 months at the time of this writing), research into its capabilities is still in the exploratory stages. The value of an LLM like OpenAI's most recent model GPT-4 is its versatility across domains, allowing it to support a wide range of tasks (Ray, 2023). Applications for GPT have been demonstrated in many industries such as healthcare, business and finance, legal services, content generation, education, programming, sales, banking and more (Ray, 2023). Essentially, the flexibility of GPT affords developers the opportunity to create tailored applications for their needs (Ollivier et al, 2023).

Implementation of GPT in many fields has faced challenges. Examples can be found in the news of ChatGPT being used clumsily, leading to failures such as erroneous periodical publications and factually incorrect legal arguments (Farhi, 2023; Bohanon, 2023). These failures stem from overreliance on AI, where individuals prompting the AI neglect to review the generated results. Instead of replacing humans, AI tools like GPT should be utilized as assistants or expert advisors, handling challenging or tedious tasks, and providing valuable insights.

GPT and other LLMs have been highlighted in many fields, one domain that was not pursued in early reviews of the technology is systems engineering (Ray, 2023). Large Language Models work by learning the broad patterns and relationships that exist within language, enabling them to comprehend the meaning of prompts, sentences, and phrases. LLMs can comprehend complex relationships and draw meaning from that complexity. Given systems engineering's emphasis on managing complexity, LLMs hold great potential to support systems engineering activities and processes. One of particular interest is in the generation of Systems Modeling Language (SysML) conceptual models.

While attention has been dedicated to assessing GPT's proficiency in generating syntactically correct programming code (Biswas, 2023; Liu et al., 2023; Sakib et al., 2023), there has been limited exploration of its capacity to develop comprehensive conceptual models and system models. This study is one of the initial endeavors to examine GPT-4's ability to generate SysML conceptual models. This research integrates GPT-4 into enterprise-level SysML modeling software (CATIA Magic) using a newly created tool, AI-Systems Modeling Enhancer (AI-SME). Moreover, this research marks an effort to evaluate GPT's performance in system modeling to that of a human. Rather than replacing systems engineers, this AI-SME tool is envisioned as an intelligent assistant that can help modelers create more complete and accurate SysML models faster than historically possible.

Artificial Intelligence, Large Language Models, and GPTs

Artificial intelligence (AI) is defined as 'a field of science and engineering concerned with the computational understanding of what is commonly called intelligent behavior, and with the creation of artifacts that exhibit such behavior (Shapiro, 1992). The focus of this research, LLMs, are a type of AI that have been trained on massive amounts of text data. This training enables the LLM to understand and generate human language and is accomplished through a combination of neural networks and machine learning algorithms. These LLMs allow computers to understand and generate language more quickly and accurately than was previously possible (Ray, 2023).

The text data, in the form of news articles, books, social media posts, and even Wikipedia, provides the LLM with a body of information that teaches the model patterns and relationships within a language. This understanding of linguistic patterns and relationships enables LLMs to generate natural responses, text, and images.

Perhaps the most well known example of an LLM is OpenAI's GPT and its accompanying chatbot app, ChatGPT. These Generative Pre-trained Transformers (GPTs) stem from Natural Language Processing AI and are built around Transformer Architecture (Ray, 2023). This revolutionary architecture paved the way for a series of OpenAI products GPT-2, GPT-3, and, most recently, GPT-4. With each iteration the number of parameters used for training the model has grown in orders of magnitude, from 1.5 billion parameters in GPT-2 to 100 trillion parameters in GPT-4. This growth led to impressive improvements in performance, for example, GPT now outperforms most humans on the bar exam (Katz, Bommarito, Gao, & Arredondo, 2023).

Model-Based Systems Engineering

Model-Based Systems Engineering (MBSE) combines the principles and practices of Systems Engineering with Model-Based methods, languages, and tools. Value-added deployments of MBSE generate representations of behavior, structure, or both that communicate accurate information to stakeholders. Regardless of the target of the MBSE analysis, it is imperative that the model is focused in purpose and scope. The application of MBSE may be focused on describing a system of interest, analyzing an element of a system, characterizing a process, simulating impacts of phenomena, or other purposes (DoD, 1998). The models that result from these efforts may represent the entirety of a system or elements of a system or process that bear particular relevance to the task at hand (Freidenthal, et al. 2014). Because MBSE has strong roots in traditional systems engineering, it can be applied broadly and liberally across industries.

Exploring Human-AI Teaming for Systems Modeling

There is a growing desire to leverage the collaboration between humans and AI, known as *human-AI teaming*. Recent developments in AI provide the opportunity for integration into user-facing systems across industries and scales of complexity (Amershi et al, 2019). However, challenges arise from the opacity of deep learning AI models, often termed *black boxes* due to their inability to explain decision-making processes to human collaborators. This lack of transparency significantly impacts the trust human-users place in their AI counterparts. Increased research in developing collaborative mental models between humans and AI becomes pivotal in fostering effective teamwork and trust (Kaur, 2019).

Zhang et. al. investigated AI assistance in multiplayer games, demonstrating humans prioritize specific characteristics in their AI teammates. Skill was deemed the most important aspect humans valued from AI, followed by supportive, proactive, and obedient traits. The human expectations from AI teammates include skill, shared understanding, sophisticated communication for collaboration, and human-like performance (2021). These guidelines can be applied to human-AI teaming in the MBSE domain. If AI can provide these characteristics in a MBSE setting, human-AI teaming will provide a valuable resource for the future.

Further investigation into the strengths and weaknesses of both humans and AI is essential. AI excels at drawing inferences from training data and creating structured model relationships with unmatched speed and knowledge. While AI performance can surpass human capabilities in some respects, gaps exist if the training data lacks relevant information associated with the conceptual model (Amershi et al, 2019). The communicative abilities of AI continue to advance, demonstrating a growing responsiveness to human instructions—an aspect highly valued by human modelers. One primary deficit pertains to shared mental models between the human and AI. Despite replicating human-like responses, AI lacks conceptual understanding and intricate mental models, as its neural networks focus on pattern recognition and may not fully capture the complexity of human thought and creative processes (Kaur, 2019).

This potential limitation could lead to frustration if the AI consistently misinterprets the human modeler's intent. Consequently, the AI might also generate incorrect modeling elements. In-order to use AI effectively in the modeling process, engineers must have training in prompting and guiding the AI. The AI outputs still need to be edited to ensure they are acceptable designs. By developing a set of parameters, engineers can harness AI and accelerate their process (Yüksel et al, 2023). Fortunately, humans retain the authority to make final decisions in the model. The modeler can audit any model elements generated by the AI, fostering a symbiotic relationship. AI excels in rapid model generation, while humans contribute by editing and adding complex relationships or associations beyond the AI's capability. This collaborative approach maximizes the strengths of both participants in the MBSE domain.

Existing Endeavors in AI and Systems Modeling

As AI capability has advanced, researchers made strides in harnessing its power to generate system models. There have been devoted efforts to developing systems that leverage NLP for the automatic generation of UML (Unified Modeling Language) diagrams (Deeptimahanti & Babar, 2009; Gulia & Choudhury, 2016; More & Phalnikar, 2012). These methods have also been extended to SysML models (Chen & Bhada, 2022). In these applications, users input textual descriptions and the algorithms then identify words of significance, such as nouns representing objects and verbs indicating actions. Using these keywords, the algorithm can generate model elements for a UML diagram (Dawood, 2017). It is important to note that the generated models are restricted to the data provided by the user and the effectiveness is reliant on the sophistication of the algorithm. Challenges exist due to the inherent ambiguities of natural language.

Relatedly, a UML diagram recommender plugin was developed using supervised learning techniques for the open-source modeling tool Eclipse Papyrus. This tool, utilizing around 100,000 models from public repositories, employs algorithms to suggest potential related elements when a specific element name is present in a model. The utilization of a Papyrus plugin allows access to AI data to generate model elements (Savary-Leblanc et al., 2021). While not utilizing generative modeling techniques, this research represents a breakthrough in the integration of AI in widely used modeling software.

On November 30, 2022, OpenAI publicly launched the ChatGPT (Chat Generative Pretrained Transformer) application. This pretrained model, based on an extensive dataset, excels in natural language processing and generates unique responses without relying on predefined answers. This technology has significantly reduced obstacles for researchers seeking to leverage AI without a specialized background in AI systems. Due to its accessibility, this tool has been adopted across industries. However, currently a limited number of research initiatives are exploring the use of the ChatGPT application and underlying GPT model for generating conceptual and systems models.

In one research study, ChatGPT's accuracy was assessed in generating UML models with PlantUML, a tool for creating diagrams from plain text. Results showed ChatGPT is proficient with simple associations, but it encounters difficulties with complex relationships. Although ChatGPT can produce models quickly and efficiently, researchers note that ChatGPT is not yet a reliable tool to perform modeling tasks. However, the study details the promise of ChatGPT being used as a modeling assistant in the future (Cámara et al., 2023). This research lays the foundation for the exploration of AI modeling assistants. Additionally, GPT's integration into enterprise-level modeling software remains an avenue for exploration.

This research aims to integrate GPT into CATIA Magic. Using GPT, users can express their modeling objectives in natural language, translating into automatic model element generation. This offers a novel and efficient approach to systems modeling. By harnessing the synergy between advanced AI models and the human modeler, this research opens avenues for enhanced user experience and streamlined workflows.

Methods

Integrating ChatGPT into MBSE (AI-SME)

CATIA Magic incorporates plugin capability that allows developers to interface with modeling tools programmatically. This enables the automation of repetitive tasks without using the regular point-and-click interface. However, writing code for plugins comes with challenges.

One significant hurdle is that MBSE involves capturing architecture of complex systems, human decision-making skills, and knowledge of similar existing systems. Trying to implement this human capability into a structured code can be quite challenging. While plugin capability exists for automation, the learning curve for implementation and the difficulties in emulating human modeling behavior through code is less practical for everyday users.

The rising popularity of AI, especially in generative AI with natural language processing, leads to an opportunity to address limitations in coding *human-like* capabilities. Integrating AI into systems modeling software through natural language prompts allows the user to create complex conceptual models harnessing extensive training data, pattern recognition, inference, and contextual understanding. This AI data can be parsed and implemented programmatically using plugin capability, significantly reducing the time and effort required for MBSE tasks.

The AI-SME in this research utilizes OpenAI's GPT to allow users to express their modeling objectives in natural language form. The AI-SME takes the user input and translates the GPT response into CATIA Magic using Open Java API plugins.Open AI offers a range of models, each trained differently and boasting incremental enhancements. This research employed the latest model (as of late November 2023), GPT-4 Turbo. Trained on data up to April 2023, this is the most up-to-date model, though it exhibits slower response times and is not currently recommended for production-level traffic. In this specific application, the flexibility provided by the Open Java API plugin enables the integration of GPT to the modeling environment. This synergy allows GPT responses to generate modeling elements with minimal translation programming, offering a powerful tool for modelers.

Data Creation Procedure

GPT shows significant response variability, with small changes in prompts resulting in distinct differences in generated responses. The inherent non-deterministic nature of the AI system produces diverse responses, making responses unpredictable even with identical prompts. To address this variability of responses, three separate prompts were crafted, each varying in specificity of the desired model output. These prompts were replicated three times, resulting in a total of nine models for assessment. Additionally, the time taken to create each diagram type in the model using the AI-SME was programmatically recorded.

The prompt sent to GPT consists of two segments. The initial segment is the user input, wherein the user provides instructions for the desired model. Subsequently, the AI-SME appends a second segment to the user input, incorporating a brief message that designates the desired SysML diagram and outlines a structure format for the GPT response. This response format ensures the AI-SME can parse the data from GPT to generate model elements. The existing graphical user interface is displayed in Figure 1, accompanied by an illustrative prompt provided below.

<u>Prompt Segment 1 (User input):</u>

Design a system for a model rocket.

Mission Statement:
The system will deliver a payload of up to 200 grams to an altitude of 1500 feet five seconds or less. Once the target altitude has been achieved, the system will return itself and the payload safely to earth.

Stakeholder desires:
The target system is self-propelled and will expediently transport the intended payload to an altitude of 1500 feet. The payload needs to be transported quickly and can sustain an inertial force of up to 30g and transient inertial force of up to 40g. The payload must have direct access to the local environment to enable measurements of environmental conditions during the flight. The payload is self-powered and exists within an envelope of 1.5 inches x 3 inches x 0.7 inches. The system will protect the payload from inertial forces that exceed the payload's capacity and will isolate the payload from shock and vibration. The system will achieve a vertical flight, such that it will maintain an attitude no more than 2 degrees from vertical drift in the presence of a 15 mph cross breeze. The system should be reusable.

<u>Prompt Segment 2 (Example of appended AI-SME input for a block definition diagram):</u>

Create a SysML block definition diagram for the system listed. In this format desired, each key should represent a component, and the corresponding values should indicate the components that interface with the respective component. Organize the hierarchy in tiered subsystems. Decompose the system into as many components as possible. For example, the format should strictly adhere to the JSON file following:

{"Component A": ["Component B", "Component E", "Component D"],
"Component B": ["Component Z", "Component B"],
"Component C": ["Component Y", "Component A"]}

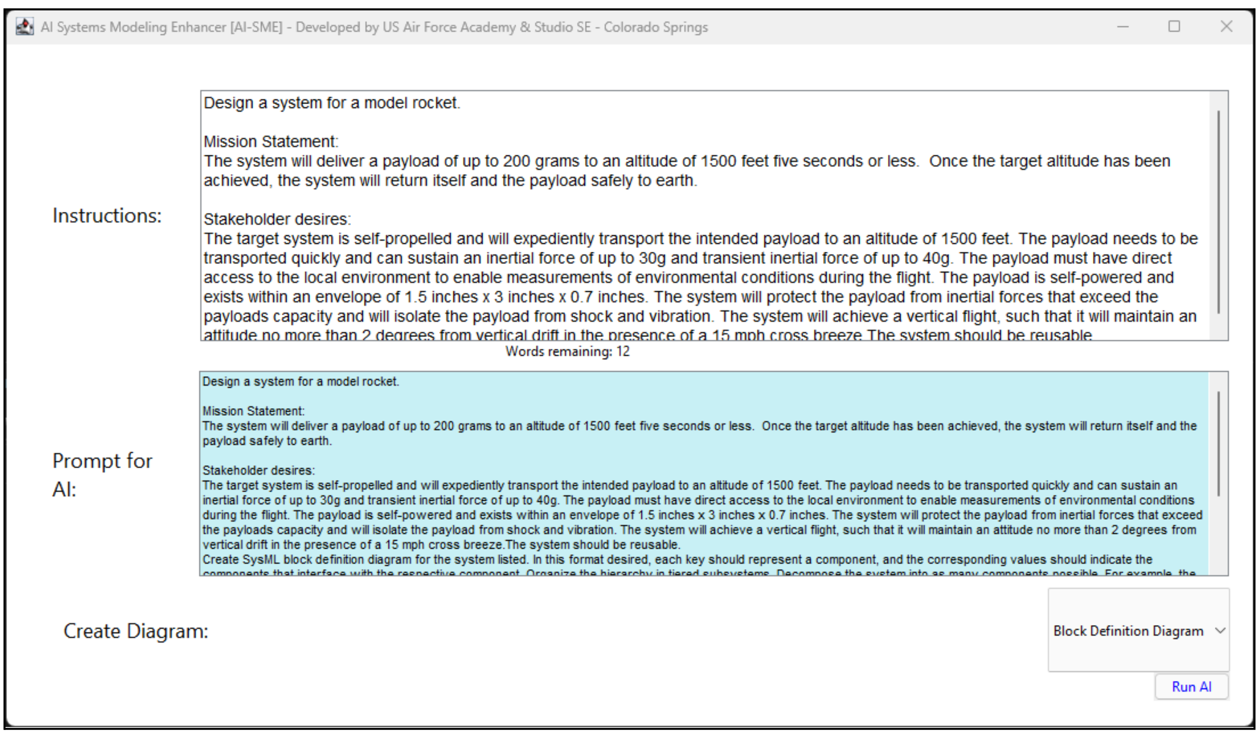

Figure 1. Example of the AI-SME plugin interface.

Evaluation of Conceptual Models

Creating a system model is an art, placing the quality burden on the human modelers. Moody states, “there are no generally accepted guidelines for evaluating the quality of conceptual models, and little agreement among experts as to what makes a ‘good’ model” (Moody, 2005). The responsibility lies with the modeler to strike a balance - communicating system elements clearly for implementation while maintaining simplicity for adequate comprehension. The model must provide valuable data to both the model creator and the model viewer.

Several attempts have been made to automatically evaluate diagrams by initially decomposing the models into units, including entities and relationships. These efforts involve employing an automated assessment process, as demonstrated in studies by Ali et al. (2007), Thomas et al. (2008), and Waugh et al. (2004). Although these techniques use quantitative metrics for assessing models, the comparison tool must have a completed solution for reference and relies heavily on the subjectivity of what the expert considers to be correct and complete. Cherfi et. al. defines the metrics for conceptual model quality into four major categories: legibility, expressiveness, simplicity, and correctness. In this study, each category is quantified as a ratio based on specific element types (entities, relations, inheritance links, etc.) (Cherfi et al., 2002). Similarly, the qualitative evaluation of the models discussed in this paper was guided by the following criteria:

1) Cohesiveness - does the model present a unified description of the system of interest?
2) Coherence - are the model elements connected in a way that is understandable and logical?
3) Completeness - does the model describe a system that could implement the mission and have all the necessary elements been created?

Results, Observations, and Discussion

AI-SME Generated Models

The AI-SME plugin was used to generate several sets of models. The AI-SME was successful in creating requirements, structural elements (blocks and part properties), and identifying interfaces among the part properties. For brevity, the results are summarized using a representative set of diagrams. AI-SME was used to generate requirements, structure, and interfaces for each of three different prompts. These prompts resulted in differing architectures and requirements, as shown in the following figures. The three prompts are summarized here:

- Prompt 1: Mission and stakeholder desires were provided as prompts to the AI.
- Prompt 2: Mission and stakeholder desires, along with the system's name, were used as prompts to the AI.
- Prompt 3: Only the name of the system was used as a prompt to the AI.

Structural Development. Figure 2 shows one example of a block definition diagram (BDD) that was generated using the Prompt 2 also shown in the AI-SME interface in Figure 1. An examination of the block definition diagram captured in Figure 2 reveals AI-SME model generation mistakes. Specifically, the number of part properties identified on the diagram exceeds the number of actual parts created. This reveals a limitation in GPT's capacity to articulate a complete hierarchical structure. Addressing this challenge could involve a two-step approach: enhancing the precision of prompts given to GPT and implementing corrections during the diagram generation. This two-fold process would ensure that each part property is associated with a corresponding part in the diagram.

While human intervention would be necessary to ensure proper deployment of the SysML language, the AI-generated structure did create levels of hierarchy of increasing specificity showing that AI-SME has the ability to recognize structural hierarchy. Additionally, AI-SME maintained a reasonable focus on structural description and did not tend to mix logical descriptions into the structural hierarchy.

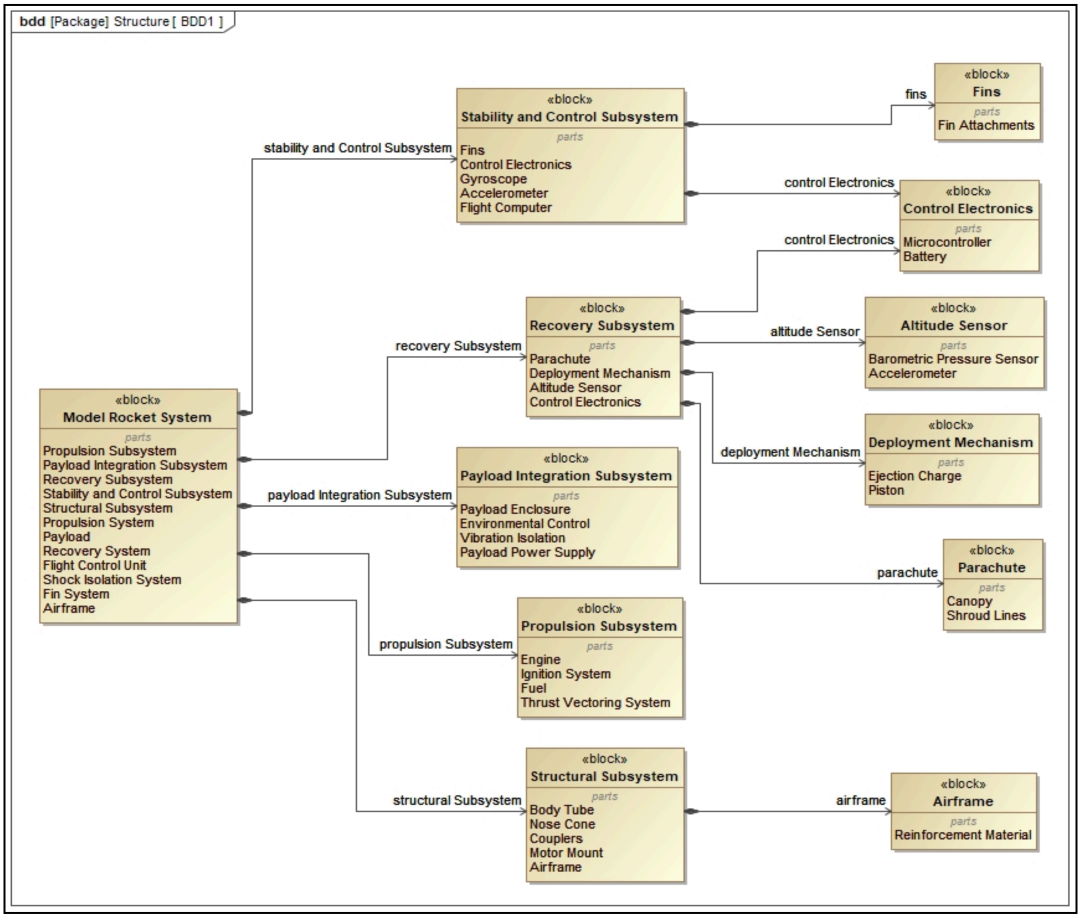

Figure 2. Example of AI-generated structural hierarchy captured in SysML Block Definition Diagram

The following points illustrate the types of mistakes found in the structure created by AI-SME. Close inspection of Figure 2 reveals AI created a part property named *Fin System* owned by the block Model Rocket System; it also created a part property named Stability and Control Subsystem which is composed of Fins. This mistake is repeated in several other instances. For example, Model Rocket System owns parts called *Airframe* and *Structural Subsystem*. Airframe is also a part property of Structural Subsystem.

AI-SME does not yet recognize the redundancy in its description nor does it maintain cohesiveness in the results it produces. Further inspection of Figure 2 shows that AI-SME created both Recovery Subsystem and Recovery System part properties for the Model Rocket System, but only decomposed the Recovery Subsystem. Additionally, AI-SME generated both Flight Control Unit and Stability and Control Subsystem part properties for the Model Rocket System, but only decomposed the Stability and Control Subsystem which is composed of Control Electronics.

AI-SME did assign parts to the Model Rocket System and generate blocks for most of the parts. It did not generate blocks for the Payload, Recovery System, Flight Control Unit, Shock Isolation System, and Fin System, even though part properties were created. AI-SME was successful in organizing the blocks it created into a hierarchy.

AI-SME generation of the structural hierarchy leaves the following tasks to the modeler:

- Remove redundant properties
- Ensure all part properties are typed by a block
- Remove redundancies
- Review to ensure cohesiveness among structural definitions

Evaluation Criteria

Cohesiveness: The structural descriptions created by AI-SME did reasonably exhibit unified representations of the system of interest. The block definition diagrams did show an ordered hierarchy; although, there were redundancies and minor inconsistencies among the layers of the hierarchy.

Coherence: The block definition diagrams created from AI-SME output were understandable and presented a set of logically-connected elements.

Completeness: AI-SME produced different results with each prompt and within each iteration of the same prompt. As the output displayed large variability, the structural definition cannot be assumed to be complete without human review

Requirements Development. Figure 3 shows a sample of requirements created by AI-SME for Prompt 2. The requirements are effectively a recitation of the mission statement and stakeholder desires. AI-SME used the word "must" rather than "shall" to denote requirements. Observations were noted that show AI-SME will alternate between "must" and "shall." There is no evidence of a behavioral analysis or risk analysis present in the example set of requirements. Two sets of requirements did identify one safety concern, but safety was not present across all sets. Additional research is needed to determine if identifying necessary behaviors as part of the prompt would result in a more complete and robust set of requirements. Such work was out of scope for this effort.

AI-SME generation of requirements leaves the following tasks to the modeler:

- Ensure requirement text, grammar, and form are consistent and conform to good requirements practices
- Review behavioral analyses to ensure requirements completely capture necessary performance and functionality of the system
- Review risk and safety analyses to ensure requirements meet safety standards and appropriate risk mitigations are in place
- Ensure requirements are consistent
- Add all traceability relationships

Evaluation Criteria

Cohesiveness: The requirements produced by AI-SME were cohesive insofar as they conformed to and reflected the information included in the prompt. Much more analysis would be necessary to ensure the requirements exhibited sufficient cohesiveness across the intended mission of the system.

Coherence: The requirements were generally found to be understandable and actionable.

Completeness: Requirements produced by AI-SME were incomplete. Although GPT was prompted with the system description of 'Model Rocket', it failed to recognize the safety requirements that are publicly available such as those in the National Association of Model Rocketry Model Rocket Safety Code.

#	Name	Text
1	1 Safe Return Requirement	Post-mission, the system must safeguard the payload and preserve the structural integrity of the system for future use.
2	1.2 Descent Control	The system must return the payload to the ground safely following payload deployment.
3	1.1 Reusability	The system components used for ascent and descent must be recoverable and reusable.
4	2 Payload Delivery Requirement	The system must efficiently transport a specified payload to a requisite altitude within a fixed timeframe.
5	2.2 Payload Capacity	The system must accommodate a payload of up to 200 grams.
6	2.3 Altitude Achievement	The system must be capable of reaching an altitude of 1500 feet.
7	2.1 Time Constraint	The payload must reach the target altitude within five seconds of launch.
8	3 Design Constraint Requirement	The system must adhere to strict physical and performance constraints as directed by the mission parameters.
9	3.1 Payload Dimension	The system must accommodate a payload with maximum dimensions of 1.5 inches x 3 inches x 0.7 inches.
10	3.2 System Propulsion	The system must be self-propelled to ensure mission objectives are met.
11	3.3 Flight Attitude Control	The rocket must maintain a vertical flight path within 2 degrees of deviation despite a 15 mph cross breeze.
12	4 Payload Protection Requirement	The operational envelope of the payload must be protected from excessive forces and environmental stresses.
13	4.2 Inertial Force Tolerance	The payload must withstand sustained inertial forces of up to 30g and transient forces up to 40g.
14	4.1 Shock Isolation	The system must isolate the payload from shock and vibration throughout the mission.
15	5 Environmental Interaction Requirement	The payload must have direct exposure to the local environment to facilitate scientific measurements during flight.

Figure 3. Requirements table showing examples of requirements created by AI-SME.

Interface Development. AI-SME was asked to generate interface definitions and corresponding internal block diagrams for each of the nine test models. Figure 4 shows the raw output from one of the iterations using Prompt 2. Inspection of the models revealed that the ports created are only assigned to part properties; no ports were assigned to blocks. Connectors, if present, are hidden from diagram view.

Figure 4 shows that AI-SME generated an interface on the Flight Control Unit for 'power out' and connected it to the 'electrical power in' interface on the Recovery System and provided a connector (not shown in the diagram, but included in the specification) to the 'Power In' interface on the Payload. It indicated that the Propulsion System has an 'Electrical Power Out' interface, but did not generate a connector to another part property. Because these ports are not completely connected, the output from AI-SME creates ambiguity in the interface definition. Again, more research is necessary to determine if adding behavioral analysis to the prompt would produce a better interface definition.

AI-SME generation of the internal block diagram leaves the following tasks to the modeler:

- Type all ports with appropriate model element
- Ensure all connectors are properly displayed on internal block diagrams
- Ensure all flow properties have appropriate types
- Add item flow information
- Add any missing ports to identified blocks
- Add all ports to blocks at lower levels of hierarchy

Evaluation Criteria

Cohesiveness: Interfaces defined by AI-SME displayed moderate to low cohesiveness as the connections they described were not explicitly defined. While the ports were given reasonably descriptive names, the directions were not always labeled nor was the intention or purpose of the ports identified.

Coherence: The interface definitions and internal block diagrams exhibited low coherence in that the relationships, when defined, were not displayed on the diagram and the ports were not typed such that the objects, data, etc. flowing through them was understood. Given the some ambiguity in port names and the fact that no information describing the motivation for creating each port was given by AI-SME, the modeler must exert considerable effort to properly connect the ports.

Completeness: The interface definitions provided by AI-SME provide a starting point for the modeler but cannot be characterized as complete. The missing port types and missing connections will require effort on the part of the modeler to describe accurately.

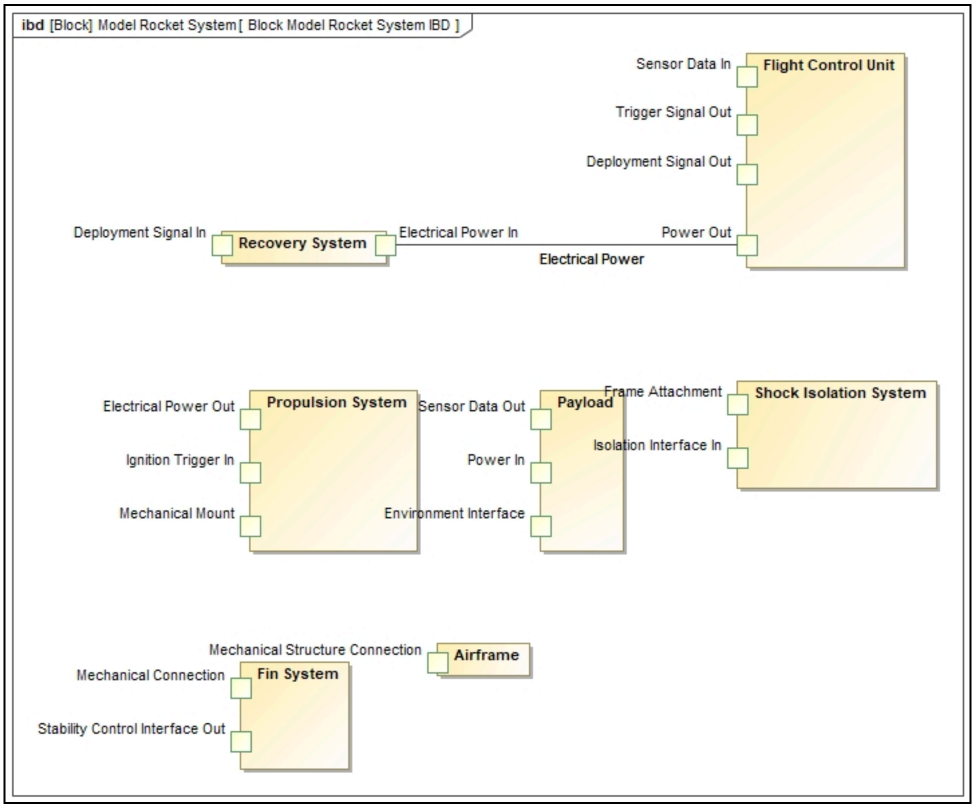

Figure 4. Example of internal block diagram created by AI-SME

Summary of AI-SME-generated models. Table 1 provides a summary of statistics related to the use of AI-SME to generate SysML models. The columns describing the number of elements produced illustrate the variability in GPT's responses. The variability among prompts is not surprising as each of the three prompts communicated different information. The variability in the responses to a single prompt is indicative of the need for effective review of the results produced by AI-SME. The columns showing time required to produce results and create SysML representations is remarkable in the context of creating an AI-powered modeling assistant.

Table 1. Summary of AI-SME data.

		Number of Elements			Time to Generate Diagrams (seconds)			
Prompt	Run	Requirements	Blocks	Ports	Requirement	Block Defition	Internal Block	Total Time
1	1	15	51	18	27.5	21	34	82.5
	2	13	59	19	30.1	26.7	47	103.8
	3	16	57	27	37.5	40.1	44.9	122.5
2	4	15	51	17	33.9	53.2	33.8	120.9
	5	18	54	17	28.7	23.1	34.7	86.5
	6	15	98	17	25.9	30.1	28.2	84.2
3	7	18	26	19	48.8	19.8	30.2	98.8
	8	20	31	14	38.6	34	32.7	105.3
	9	20	25	13	31.9	14.1	45.7	91.7
							Average Time	99.6

Human-Generated Model

Figures 5, 6, and 7 show diagrams from a human-created model. The human-created model was generated following a process such as that outlined in (Author, 2015). The human-generated model was produced from the same *Prompt Segment 1 (User Input)* described above. The human modeler had no knowledge of the output of AI-SME while creating the model. The human-created model was developed to compare and contrast the output of AI-SME with the output of a modeling effort performed by a Systems Engineer. It is important to note that the human modeler had no prior experience modeling, building, or flying model rockets. The human modeler and the AI-SME operator received the information in *Prompt Segment 1 (User Input)* concurrently. Neither the human modeler nor the AI-SME operator were given any advantage in the effort.

#	Name	Text	Refined By	Derived From
1	15 Payload Mass	The system shall support a payload of up to 200 grams.		SN-47 Mission Statement
2	16 Maximum Altitude	The system shall transport a payload to an altitude of less than or equal to 1500 feet.		SN-40 Self Propelling System SN-47 Mission Statement
3	17 Flight Time	The system shall reach its maximum altitude in less than or equal to 5 seconds.		
4	18 System Recovery	Upon reaching the target altitude, the system shall return itself to Earth.		SN-47 Mission Statement
5	19 Attitude in Cross Wind	The system shall maintain an attitude of at most 2 degrees from vertical lift in the presence of a 15 mph cross wind.	78 Stability Margin 85 Tolerance of Flight Pa	SN-45 Vertical Flight
6	20 Payload Accommodation	The system shall support a payload with dimensions of 1.5 inches x 3 inches x 0.7 inches.		SN-43 Payload Dimensions
7	21 Payload Environment Interface	The system shall enable the payload to interface directly with the external environment.		SN-42 Environmental Monitoring
8	48 Self Propelled	The system shall be self-propelled.		SN-40 Self Propelling System
9	49 Payload Recovery	Upon reaching the target altitude, the system shall return a payload to Earth.		SN-47 Mission Statement
10	52 Total Mass	The total mass of the system shall be less than or equal to 1500 grams.		Std-1 Gross Launching Mass 17 Flight Time
11	65 Environmental Harm	The system shall prevent harm to the environment.		
12	77 Center of Gravity	The center of gravity of the system shall be in front of the center of pressure.		19 Attitude in Cross Wind
13	78 Stability Margin	The system shall have a stability margin of at least [x].		
14	81 Mechanism to Maintain Stability	The system shall have a mechanism to maintain stability during flight.		19 Attitude in Cross Wind
15	85 Tolerance of Flight Path	The system shall follow its flight path within +/- [x] m of final landing destination.		
16	89 Wadding Min Temperature	The wadding shall withstand temperatures of at least [x] K.		51 Payload Damage
17	93 Airframe Load	The airframe shall withstand a load force of [x] kPa.		
18	95 Attachment	Upon ejection charge emission, the nose, airframe, and recovery mechanism shall remain attached.		
19	96 Shock Cord Force	The shock cord shall withstand a minimum force of [x] N.		95 Attachment
20	97 Payload Bay Attachment	Upon ejection charge emission, the payload bay shall remain in the airframe.		
21	98 Drag Force	The system shall provide a mechanism to create a drag force of less than [x] N.		52 Total Mass
22	99 Lift Force	The system shall provide a mechanism to create a lift force of at least [x] N.		52 Total Mass
23	Launch Pad and Launcher Requirements			
36	Motor Requirements			
43	Payload Bay Requirements			
50	Recovery Requirements			

Figure 5. Human generated requirements. Inclusive of system and component level requirements.

Inspection of the sample of requirements in Figure 5 reveals multiple categories of requirements were identified. The requirements have been decomposed to several levels of hierarchy based on a robust systems engineering analysis of the system. The human-created requirements include function, non-functional, and safety/risk-related requirements. Additionally, the human modeler has added traceability information to the model. In contrast, AI generated the requirements strictly from the mission statement without further elaboration or recognition of any additional requirements for a Model Rocket.

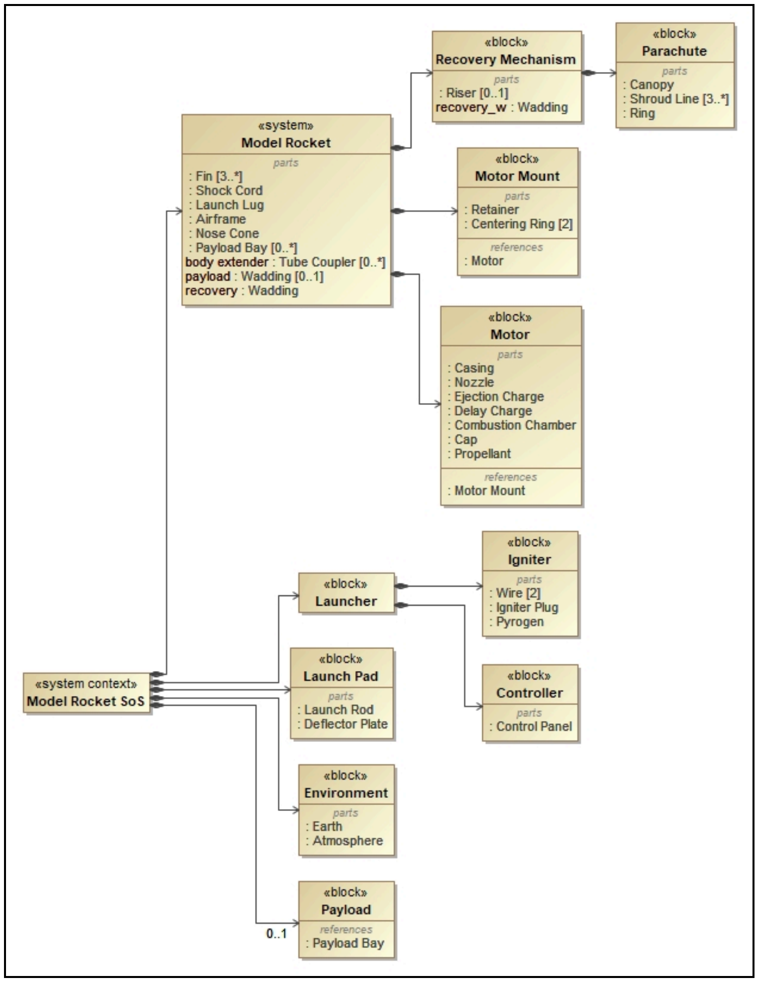

Figure 6. Example of block definition diagram created by human modeler.

Figure 6 demonstrates a block definition diagram created by the human modeler that illustrates the impact of following a good systems engineering process in the development of models and model elements. The blocks were created based on the outcome of the systems engineering work and are deliberately created to satisfy requirements and achieve behaviors deemed necessary by the analysis. Part properties are properly typed and there are no redundancies present.

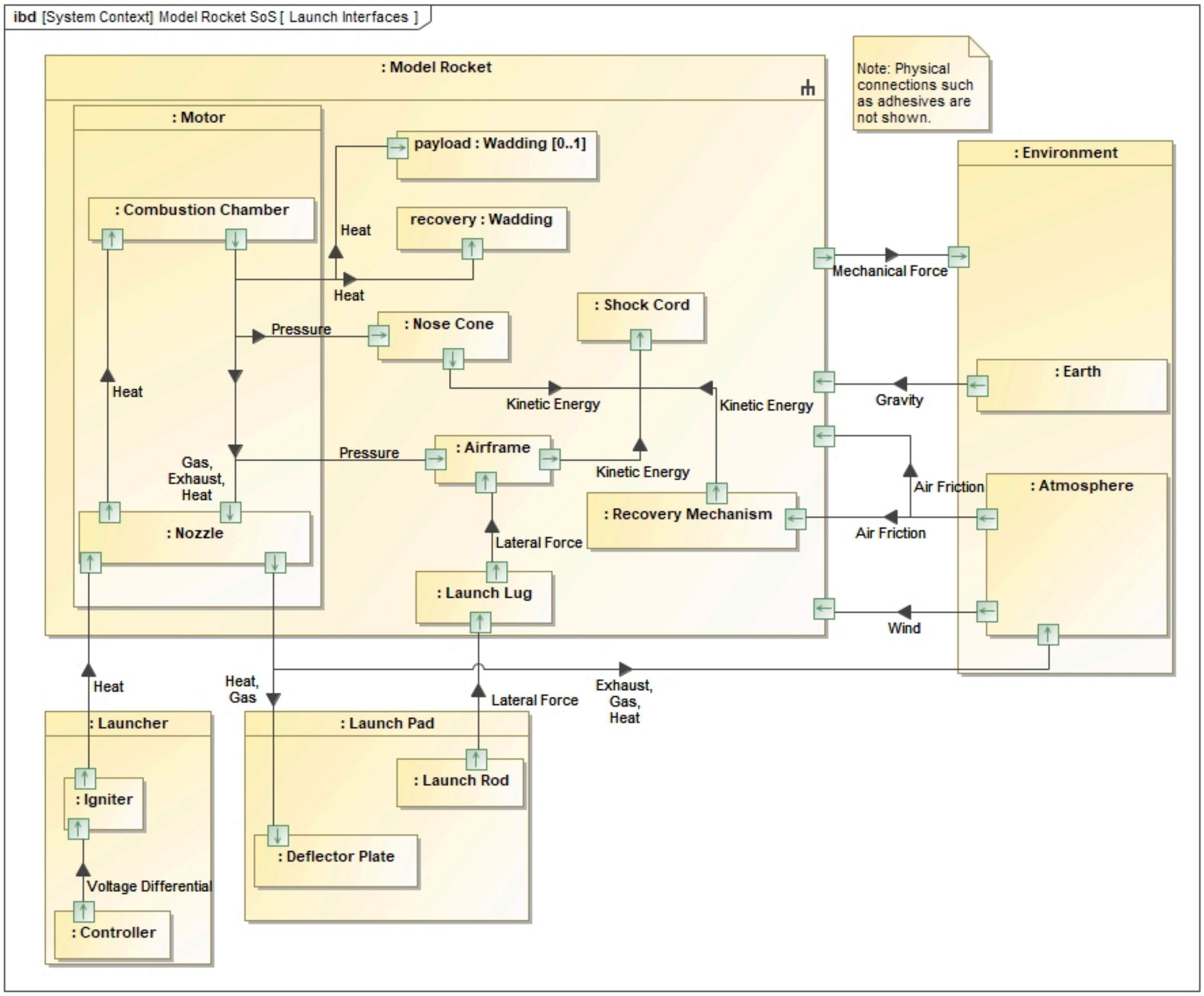

Figure 7. Internal block diagram created by human modeler.

Figure 7 shows an internal block diagram created by the human modeler. Upon inspection of the diagram, it is apparent that all ports were deliberately assigned and properly typed. The direction of flow is noted and connectors convey item flow. The ports were assigned to blocks such that any part properties typed by those blocks would properly implement the identified ports.

Overall Human Modeler Evaluation Criteria

Cohesiveness: The model developed by the human modeler was found to be highly cohesive. Evidence that a robust systems engineering process was followed is apparent in the flow and organization of the requirements, structural elements, and interface definitions.

Coherence: The human modeler produced a model that is highly coherent. Elements are connected as necessary and the model is understandable and functions as an effective communication tool.

Completeness: The model developed by the human modeler is reasonably complete and would result in the development of a system that satisfies the stated mission, the identified requirements, and the stated stakeholder desires.

It is important to note that, in contrast to AI-SME, the human modeler has extensive knowledge of the SysML language. The human modeler also has a solid understanding of the systems engineering process and corresponding analyses and methods. AI-SME relies on the unknown *experience* of GPT.

Conclusion

As one of the initial endeavors to examine GPT-4's ability to generate SysML conceptual models, this study is a significant advancement in the field of AI assisted systems engineering. The use of generative AI to create SysML diagrams was accomplished through creation of a tool that integrates GPT-4 with enterprise-level SysML modeling software (CATIA Magic). The tool took the form of an MBSE intelligent assistant, AI-SME, capable of rapidly producing requirements diagrams, block definition diagrams, and internal block diagrams. Using AI-SME, a sample set of these diagrams was created for a hypothetical model rocket system while a second set of models was created by a human modeler.

Through a comparison of the human and AI generated model sets we found that AI-SME and the human modeler developed comprehensive structural hierarchies that were capable of executing the mission. However, the redundancies and inconsistencies present in the AI-generated models require analysis, review, and revision to ensure coherence and cohesiveness in the description. Additionally, AI-generated models developed highly varied numbers of blocks among iterations and among prompts. This variance would need to be tempered by prudent application of systems engineering principles and the use of good systems architecting principles to create an optimal structural description.

It was found that the requirements created by AI-SME were coherent and cohesive but were incomplete and often were a simple recitation of the information included in the prompt. More research is necessary to determine if the completeness of the requirements could be improved by more detailed prompts.

The interfaces defined by AI-SME were moderately coherent and cohesive, but were not completely defined. Similarly to requirements, more research is needed to determine how to better enable GPT to create interface descriptions that are accurate for the system of interest.

Overall, AI-SME required 99.6 seconds to create requirements, structural definitions, and interface definitions. This represents a considerable time savings and could offer a distinct advantage to a systems modeling effort. In contrast, the time required for the human modeler to create the same three types of information was around 10 hours. This illustrates the potential for AI-SME to greatly reduce the time required to model a system of interest.

Especially in the case of requirements and interface development, it is not fair to denounce AI-SME as inadequate compared to a human modeler. The human modeler has extensive knowledge of systems engineering, systems thinking, and the SysML language. The comparison is interesting, however, in that it illustrates the possibility that GPT and AI-SME could evolve to be powerful allies to the human modeler.

Despite some inaccuracies in the AI generated models, AI-SME offers the potential to greatly improve the systems engineer modeling approach. This is particularly true if the AI-SME is harnessed as an assistant rather than a human modeler replacement. This work demonstrates AI-SME is viable as a modeling assistant to automate primary tasks and increase efficiency by developing prototype architectural descriptions. The human modeler can then easily review and modify the results to ensure cohesiveness, consistency, and completeness by applying a robust and thorough systems engineering analysis.

In 2020, the National Defense Industrial Association Systems Engineering Division (NDIA-SED), the International Council on Systems Engineering (INCOSE), and Systems Engineering Research Center (SERC) released a collaborative report highlighting the maturity of MBSE across government, industry, and academia. When surveying 168 enterprise participants, aggregated results show that respondents agree that “Modeling activities in our organization provide measurable improvements within and across projects.” However, of all the inquiries posed to survey respondents, the strongest disagreement stemmed from the statement “We have sufficient staffing in our organization to fill all MBSE-related roles” (McDermott et al., 2020). This indicates there are benefits from embracing MBSE, but issues stem from the lack of staffing to support MBSE. The AI-SME accelerates the modeling process, alleviating staffing issues, while improving overall project outcomes and operational efficiency.

Recommendations for Future Work

Artificial intelligence assisted MBSE is still in its infancy. This study illustrated the feasibility of integrating AI with enterprise-level modeling software, yet significant development is required before a tool like the AI-SME achieves complete maturity. First, the tool's capabilities must be expanded to encompass the generation of all 9 SysML diagrams. As the tool’s functionality grows, it is imperative to address the evaluation of SysML diagrams. Currently, there is no validated framework for assessing the quality of SysML diagrams, highlighting the crucial need for a framework to understand the value that AI-assisted MBSE can provide.

Recommendations

1. Expand AI-SME capabilities to model all SysML diagrams
2. Explore how evaluators assess models build by AI vs Human
3. Develop a framework for assessing SysML diagram quality
4. Define the strengths and weaknesses of AI-SME so that humans can most effectively employ the tool when creating models
5. Explore how to train the model in order to generate higher quality output

Not surprisingly, many systems engineers have voiced skepticism about the utility of AI for MBSE (Boozingislosing, 2023). This opposition of AI within MBSE can be explained by individual disposition toward AI, situational factors, and learned factors based on performance. Future research should explore how these factors can be addressed to ensure appropriate implementation of AI in systems engineering. Along these lines, further research should investigate how insights into the AI’s strengths and weaknesses in systems engineering can be leveraged to ensure human-AI teaming is optimized. A final avenue for future research is on training and model refinement for specific engineering domains enabling higher quality outputs from the GPT model.

Assessing the quality of AI-generated SysML diagrams remains challenging due to the absence of agreed-upon metrics for a *good* model. While standards exist across the discipline of engineering, establishing quality benchmarks for conceptual modeling is an ongoing effort and debate. As mentioned previously, clumsy implementation of AI can lead to costly errors and the AI-SME is not immune to errors. Currently, the AI-SME can generate SysML block definition diagrams, internal block diagrams, and requirement diagrams. Additionally, there is potential in exploring GPT’s ability to generate more complex and nuanced SysML relationships between elements.

References

Ahmad, A., Waseem, M., Liang, P., Fahmideh, M., Aktar, M. S., & Mikkonen, T. (2023). Towards

Human-Bot Collaborative Software Architecting with ChatGPT. *Proceedings of the 27th International Conference on Evaluation and Assessment in Software Engineering*, 279–285. https://doi.org/10.1145/3593434.3593468.
Ali, N. H., Shukur, Z., & Idris, S. (2007). A design of an assessment system for UML class diagram. *2007 International Conference on Computational Science and Its Applications (ICCSA 2007)*, 539–546. https://ieeexplore.ieee.org/abstract/document/4301193/.
Amershi, S., et al (2019). Guidelines for Human-AI Interaction. *Conference on Human Factors in Computing Systems.* https://doi.org/10.1145/3290605.3300233.
Batra, D., & Antony, S. R. (2001). Consulting support during conceptual database design in the presence of redundancy in requirements specifications: An empirical study. *International Journal of Human-Computer Studies*, *54*(1), 25–51.
Biswas, S. (2023). Role of ChatGPT in Computer Programming: ChatGPT in Computer Programming. *Mesopotamian Journal of Computer Science*, *2023*, 8–16. https://doi.org/10.58496/MJCSC/2023/002.
Bohanon, M. (2023, June 8). *Lawyer Used ChatGPT In Court—And Cited Fake Cases. A Judge Is Considering Sanctions*. Forbes. https://www.forbes.com/sites/mollybohannon/2023/06/08/lawyer-used-chatgpt-in-court-and-cited-fake-cases-a-judge-is-considering-sanctions/?sh=618c496a7c7f
Boozingislosing, "MBSE powered by AI will replace your SE role as you know it!" *Reddit*, 28 November 2023,https://www.reddit.com/r/systems_engineering/comments/10si099/mbse_powered_by_ai_will_replace_your_se_role_as/.
Cámara, J., Troya, J., Burgueño, L., & Vallecillo, A. (2023). On the assessment of generative AI in modeling tasks: An experience report with ChatGPT and UML. *Software and Systems Modeling*, *22*(3), 781–793. https://doi.org/10.1007/s10270-023-01105-5.
Chami, M., Zoghbi, C., & Bruel, J.-M. (2019). A First Step towards AI for MBSE: Generating a Part of SysML Models from Text Using AI. *A First Step towards AI*.
Chen, M., & Bhada, S. V. (2022). Converting natural language policy article into MBSE model. *INCOSE International Symposium*, *32*(S2), 73–81. https://doi.org/10.1002/iis2.12897.
Cherfi, S. S.-S., Akoka, J., & Comyn-Wattiau, I. (2002). Conceptual Modeling Quality—From EER to UML Schemas Evaluation. In S. Spaccapietra, S. T. March, & Y. Kambayashi (Eds.), *Conceptual Modeling—ER 2002* (Vol. 2503, pp. 414–428). Springer Berlin Heidelberg. https://doi.org/10.1007/3-540-45816-6_38.
Dawood, O. S. (2017). From requirements engineering to uml using natural language processing–survey study. *European Journal of Industrial Engineering*, *2*(1), pp-44.
Deeptimahanti, D. K., & Babar, M. A. (2009). An Automated Tool for Generating UML Models from Natural Language Requirements. *2009 IEEE/ACM International Conference on Automated Software Engineering*, 680–682. https://doi.org/10.1109/ASE.2009.48.
DoD. (1998).Modeling and Simulation (M&S) Glossary, DoD 5000.59-M. *Undersecretary of Defense for Acquisition Technology.*
Farhi, P. (2023, January 17). *A news site used AI to write articles. It was a journalistic disaster.* The Washington Post. https://www.washingtonpost.com/media/2023/01/17/cnet-ai-articles-journalism-corrections/
Friedenthal, S., Moore, A., Steiner, R. (2014). A Practical Guide to SysML, Third Edition: The Systems Modeling Language (3rd. ed.). *Morgan Kaufmann Publishers Inc.*
Gulia, S., & Choudhury, M. T. (2016). *An efficient automated design to generate UML diagram from Natural Language Specifications*.
Hooshmand, Y., Adamenko, D., Kunnen, S., & Köhler, P. (2018). Semi-Automatic Creation of System Models based on SysML Model Libraries. *INCOSE EMEA Sector Systems Engineering Conference*.
Katz, Daniel Martin, et al. "Gpt-4 passes the bar exam." *Available at SSRN 4389233* (2023).
Kaur, H. (2019). *Building Shared Mental Models between Humans and AI for Effective Collaboration*.

https://www.semanticscholar.org/paper/Building-Shared-Mental-Models-between-Humans-and-AI-Kaur/bc47b075dcbb2a170740d5da1d51d954efb62748.

Kazi, Z., Radulović, B., Berković, I., & Kazi, L. (2017). Ontology-based reasoning for entity-relationship data model semantic evaluation. *Technical Gazette*, *24*(Supplement 1), 39–47.

Liu, J., Xia, C. S., Wang, Y., & Zhang, L. (2023). Is your code generated by chatgpt really correct? Rigorous evaluation of large language models for code generation. *ArXiv Preprint ArXiv:2305.01210*.

McDermott, T., Hutchison, N., Clifford, M., Van Aken, E., Salado, A., & Henderson, K. (2020). *Benchmarking the Benefits and Current Maturity of Model-Based Systems Engineering across the Enterprise: Results of the MBSE Maturity Survey*.

Moody, D. L. (2005). Theoretical and practical issues in evaluating the quality of conceptual models: Current state and future directions. *Data & Knowledge Engineering*, *55*(3), 243–276.

More, P., & Phalnikar, R. (2012). Generating UML diagrams from natural language specifications. *International Journal of Applied Information Systems*, *1*(8), 19–23.

Ray, Partha Pratim. "ChatGPT: A comprehensive review on background, applications, key challenges, bias, ethics, limitations and future scope." Internet of Things and Cyber-Physical Systems (2023).

Sakib, F. A., Khan, S. H., & Karim, A. H. M. (2023). Extending the frontier of chatgpt: Code generation and debugging. *ArXiv Preprint ArXiv:2307.08260*.

Savary-Leblanc, M., Le-Pallec, X., & Gérard, S. (2021). A modeling assistant for cognifying MBSE tools. *2021 ACM/IEEE International Conference on Model Driven Engineering Languages and Systems Companion (MODELS-C)*, 630–634. https://doi.org/10.1109/MODELS-C53483.2021.00097.

Savary-Leblanc, M. (2019). Improving MBSE Tools UX with AI-Empowered Software Assistants. *2019 ACM/IEEE 22nd International Conference on Model Driven Engineering Languages and Systems Companion (MODELS-C)*, 648–652. https://doi.org/10.1109/MODELS-C.2019.00099.

Schmidhuber, J. (2015). Deep learning in neural networks: An overview. *Neural Networks*, *61*, 85–117.

Schräder, E., Bernijazov, R., Foullois, M., Hillebrand, M., Kaiser, L., & Dumitrescu, R. (2022). Examples of AI-based Assistance Systems in context of Model-Based Systems Engineering. *2022 IEEE International Symposium on Systems Engineering (ISSE)*, 1–8. https://doi.org/10.1109/ISSE54508.2022.10005487.

Shapiro, Stuart C. "Encyclopedia of artificial intelligence second edition." *New Jersey: A Wiley Interscience Publication* (1992).

Thomas, P., Smith, N., & Waugh, K. (2008). Automatically assessing graph‐based diagrams. *Learning, Media and Technology*, *33*(3), 249–267. https://doi.org/10.1080/17439880802324251.

Thomas, P., Smith, N., & Waugh, K. (2009). Automatically assessing diagrams. *Proceedings of the IADIS International Conference on E-Learning*. https://www.researchgate.net/profile/Pete-Thomas/publication/42799920_Automatically_assessing_diagrams/links/0fcfd5060076dd8ba2000000/Automatically-assessing-diagrams.pdf.

Waugh, K., Thomas, P., & Smith, N. (2004). *Toward the automated assessment of entity-relationship diagrams*. https://oro.open.ac.uk/2455/.

Yüksel, N., Börklü, H. R., Sezer, H. K., Canyurt, O. E. (2023). Review of artificial intelligence applications in engineering design perspectives. *Engineering Applications of Artificial Intelligence.* https://doi.org/10.1016/j.engappai.2022.105697.

Zhang, R., McNeese, N. J., Freeman, G., & Musick, G. (2021). "An Ideal Human": Expectations of AI Teammates in Human-AI Teaming. *Proceedings of the ACM on Human-Computer Interaction*, *4*(CSCW3), 1–25. https://doi.org/10.1145/3432945.

Author, (2015). Details withheld for review.

Diagrams created using CATIA Magic, version 19, service pack 4 by Dassault Systemes and GPT 4-Turbo (2023) with AI-SME Plug In.

Biography

Brian Johns. Brian Johns joined the United States Air Force Academy in 2023. Prior to this, he was the Chair of the Department of Physics & Engineering at Cornell College. His past engineering experiences range from building virtual-reality surgical simulators to developing mobile apps. Professor Johns has taught over twenty distinctive college courses throughout the last decade, across diverse institutions both within the United States and internationally. Outside of the classroom, he enjoys writing computer code, designing, and manufacturing.

Kristina Carroll. Kristina Carroll is a Principal Trainer and the Director of Product Development for Studio SE. She teaches advanced topics in MBSE and has an affinity for analyzing complex, cross-disciplinary problems using the Cameo System Modeler's Simulation Toolkit. Kristina has extensive experience in the medical device and cell therapy fields and has demonstrated success in using systems thinking and statistical methods for product and process development, optimization, and characterization. Kristina earned a M.E. in Biomedical Engineering from the Rutgers University and a B.S. in Chemical and Biological Engineering from the University of Colorado at Boulder

Casey Medina. Casey Medina leads Studio SE as the President and CEO. Under his leadership, Studio SE provides a range of exemplary professional development courses including Model-Based Systems Engineering. He has practiced systems and quality engineering across industries including medical device, aerospace, defense, and transportation industries. His expertise extends across disciplines including system development, requirements engineering, model-based systems engineering (MBSE), human factors, quality engineering, risk management, and medical device design control. Professionally, he is focused on developing the art of systems engineering in a manner that fosters adoption and acceptance by organizations resistant to change. He applies systems engineering practices and principles to enable rapid system development and is working to enhance the use of MBSE as an enabler for usability and human factors analyses. He has deployed MBSE to analyze social systems and homelessness. Casey and his team at Studio SE are pioneering the use of MBSE to evaluate natural systems and deploy MBSE to manage and solve sustainability issues.

Rae Lewark. Rae Lewark is an ecologist with Studio SE, focused on the integration of human and ecological systems through sustainable design. Having performed research in both the arachnid biodiversity of Ecuador and the anthropomorphic influence of ecologies in film and literature, Rae operates in the intersection between environment and human behavior. A freedive athlete and trainer, she believes strongly the reintegration of humans and the environment begins with reconnecting to nature. Her experience includes regenerative system building, circular economy design, and biomimicry-based sustainability. Rae graduated from the University of Colorado at Boulder with a degree in Environmental Studies specializing in Ecology**.**

James Walliser. Lieutenant Colonel James Walliser, Ph.D., is an Assistant Professor in the Department of Mechanical Engineering and Director of the Systems Engineering Program at the U.S. Air Force Academy. In this role he equips 200+ cadets in the Systems Engineering Major with tools and experiences necessary to solve the world's most complex socio-technical problems. He leads a team of systems engineering professors and instructors delivering a comprehensive systems engineering education including the latest in model based systems engineering. In previous roles, he oversaw engineering efforts for the nation's air delivered leg of the nuclear triad serving in multiple positions within the Long Range Standoff Cruise Missile and the B61-12 Tailkit Assembly program offices guiding these programs through successful development activities, major acquisition milestones, and numerous system engineering technical reviews. He also served as Lead F-22 Human Factors Engineer overseeing numerous human factors developmental test events. LtCol Walliser received his Ph.D. in Human Factors and Applied Cognition from George Mason University, Fairfax, Virginia.